Robust and Explainable Classical ML for Tomato Disease Recognition

Final project report. CS-871 Machine Learning, MS(AI), NUST SEECS, Fall 2026. Hamna Sadia Rizwan, Maria Ilyas Malik, Manahil Kamran. 13 November 2026.

1. Summary

Bottom line. Classical models trained on one image source do not transfer, and neither do their explanations. Trained on seven pooled sources and tested on an eighth, they recognise healthy vs diseased leaves with balanced accuracy 0.66 (hand-crafted + lesion features, logistic regression) up to 0.75 (24 CLIP concept scores from a frozen network). Six-class diagnosis transfers poorly with formula or k-means features (macro-F1 0.33) and moderately with frozen CLIP embeddings (0.52). **Feature-level explanation works:** 24 of 205 named features relate to disease in the same direction on at least 6 of 7 independent sources, led by leaf-relative yellow, non-green and lesion-size measurements and by brown/olive k-means colour words. A model's importance ranking does not recover these features.

What we did.

- Phase 1-2 (to 27 Oct): 105 hand-crafted colour, texture and shape features; RF, SVM, LR; grouped splits; 50 perturbations, augmentation, illumination normalisation, shortcut controls.
- Task 2 (12 Nov): grouped re-evaluation of the supplied 7,200-image set; cross-dataset test on PlantDoc and two Bangladeshi field sets.
- Phase 3 (12-13 Nov): an eight-source, de-duplicated benchmark (7,754 images); leave-one-source-out evaluation; two learned-extractor arms compared with the formula features, all trained and tested with classical ML: k-means visual words (classical, unsupervised) and frozen deep networks (CLIP, ResNet-50, no training); feature-level explanation with bootstrap CIs per source, glass-box models, explanation-guided selection.

Main findings.

- Related images inflate benchmarks: supplied-set accuracy 0.96 (image split) vs 0.66 (grouped).
- About a third of our raw photos are also in PlantDoc (296 of 813 by content).
- Single-source models collapse on other sources (macro-F1 0.21-0.33); pooled training restores binary transfer (0.66-0.75).
- For six-class diagnosis the frozen deep embeddings transfer clearly better than formula or k-means features (0.52 vs 0.33 macro-F1, higher on 6 of 7 sources).
- Linear models transfer better than RF and SVM; RF almost never predicts healthy on a new source (healthy recall 0.22).
- The best deep features (CLIP concepts) partly rely on capture context (plain background, hand, studio), shown by the glass-box weights.
- Leaf-relative lesion formulas are the most source-stable family (8 of 20), against 7 of 105 hand features and 1 of 24 CLIP concepts.

2. Literature review

2.1. Laboratory versus field datasets

Most image-based plant-disease work builds on **PlantVillage**, about 54,000 leaf images of 14 crops photographed one leaf at a time on a plain background [1]. Mohanty et al. [2] trained convolutional networks on it and reached 99.35% accuracy on a held-out PlantVillage test set, but only about 31% on images collected from trusted online sources under different conditions. They concluded that more diverse training data is needed. Ferentinos [3] likewise reported very high accuracy on a large laboratory-dominated collection together with a marked drop when laboratory-trained models were applied to field images. Barbedo [4] listed the factors that limit real-world use, among them covariate shift between training and deployment images and the difficulty of separating symptoms from the background. Noyan [5] showed that PlantVillage carries capture-related bias: a model using only eight background pixels classified it far above chance (49% vs 2.6%).

Field and in-the-wild datasets were built in response. **PlantDoc** [6] collected 2,598 web images of 13 species with cluttered backgrounds. **Tomato-Village** [7] photographed tomato plants in fields in Rajasthan, India, motivated by the observation that PlantVillage-trained models do not work well in real fields. **PlantWild** [8] is the largest in-the-wild collection (18,542 images, 89 classes) and stresses small differences between diseases and large variation within one disease. Regional field datasets for tomato from Pakistan [9] and Bangladesh [10]–[12] are recent and, to our knowledge, have no published benchmark. A common pattern is that each dataset is evaluated **within itself**; how a model trained on one source behaves on another is rarely reported.

2.2. Hand-crafted features and classical machine learning

Before deep learning, disease recognition relied on segmentation followed by colour, texture and shape descriptors [13]. Texture is usually described with grey-level co-occurrence statistics [14] and local binary patterns [15]; colour with channel moments and histograms. Classifiers are typically support vector machines [16] and random forests [17]. For tomato, Mokhtar et al. [18] combined Gabor wavelet features with SVMs, Hlaing and Zaw [19] modelled SIFT-based texture statistics together with colour moments, and Tan et al. [20] compared SVM, k-NN and random forests on hand-crafted features with several CNNs. Singh and Misra [21] combined segmentation with soft-computing classifiers. These studies report high accuracy, but almost all use PlantVillage images with random image-level splits, so they measure performance on a single laboratory source. Classical pipelines remain attractive for low-resource settings because they are cheap and their inputs are named, measurable quantities.

2.3. Robustness and distribution shift

Hendrycks and Dietterich [22] showed that image classifiers degrade under common corruptions such as blur, noise and brightness change, which motivates controlled perturbation tests. Torralba and Efros [23] showed that every dataset has its own "signature" and that models trained on one dataset transfer poorly to another. Domain-generalisation work evaluates by holding out whole domains; Gulrajani and Lopez-Paz [24] showed that, with a careful leave-domain-out protocol, many specialised methods do not beat plain empirical risk minimisation, and WILDS [25] collected real distribution shifts for benchmarking. Classical remedies for lighting variation include grey-world colour constancy [26] and contrast-limited adaptive histogram equalisation (CLAHE) [27]; data augmentation is the usual learning-based remedy [28].

2.4. Shortcut learning, leakage and duplicates

Geirhos et al. [29] describe shortcut learning: models exploit cues that work on the benchmark but not on the intended task. In medical imaging, models learned hospital- or device-specific signals [30], [31], and DeGrave et al. [31] showed that good external-test accuracy alone does

not prove that a model uses the pathology. Kapoor and Narayanan [32] catalogue data leakage as a main cause of irreproducible machine-learning results, including duplicates across training and test sets. Barz and Denzler [33] found that 3.3% of CIFAR-10 and 10% of CIFAR-100 test images have near-duplicates in training. Roberts et al. [34] showed that when data are grouped or structured, blocked (grouped) cross-validation is more appropriate than random splitting for predicting to new data. Label errors are a related problem; confident learning [35] flags likely mislabelled examples from out-of-sample predicted probabilities.

2.5. Explanation of models and of features

Feature-importance measures explain a fitted model. Random-forest impurity importance is biased towards some feature types [36], permutation importance can extrapolate to unrealistic inputs when features are correlated [37], and the importance of a feature depends on which of many equally good models is chosen [38]. Local explainers such as LIME [39] and SHAP [40] often disagree with each other [41]. Rudin [42] argues that for high-stakes decisions inherently interpretable models should be preferred to post-hoc explanation. In plant pathology, Toda and Okura [43] visualised a CNN trained on PlantVillage and found that it attends to lesion colour and texture, but on a single laboratory dataset.

Concept-based explanation expresses a model in human terms. TCAV [44] tests a network's sensitivity to user-defined concepts, and concept bottleneck models [45] predict through a layer of named concepts. With vision-language models such as CLIP [46], concept scores can be obtained without concept labels: Label-free CBM [47] and LaBo [48] generate concepts with language models and score them with CLIP. In this project, models are trained and tested only with classical ML; we compare two kinds of learned feature extractor. The classical one follows the bag-of-visual-words idea [52], [53]: codebooks of colour and texture prototypes learned with k-means [54], [55]. The deep one uses frozen pretrained networks, OpenCLIP [49] and an ImageNet ResNet-50 [50], purely as feature extractors.

2.6. Gap addressed by this project

(i) Classical tomato-disease studies report high accuracy on laboratory images with image-level splits; we evaluate on field photos with grouped splits and quantify how much related images inflate a benchmark. (ii) Field datasets are benchmarked within themselves; we test across eight image sources with leave-one-source-out evaluation and check dataset provenance. (iii) Explanations are usually reported for one model on one dataset; we ask which features — formula-based, learned by k-means, or extracted by a frozen deep network — keep the same effect across sources, and whether models restricted to them transfer better.

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3. Data

Eight image sources, de-duplicated across sources with a rule fixed before Phase 3 (16x16 grey thumbnails, correlation at least 0.95 under 8 rotations and flips, or identical bytes). 425 duplicates were dropped and 72 label conflicts recorded; near-duplicates within a source are grouped and never split. Excluded: the supplied Augmented Dataset (derived from our photos) and the Dinajpur set (its labels mark lesions, and class ids could not be matched).

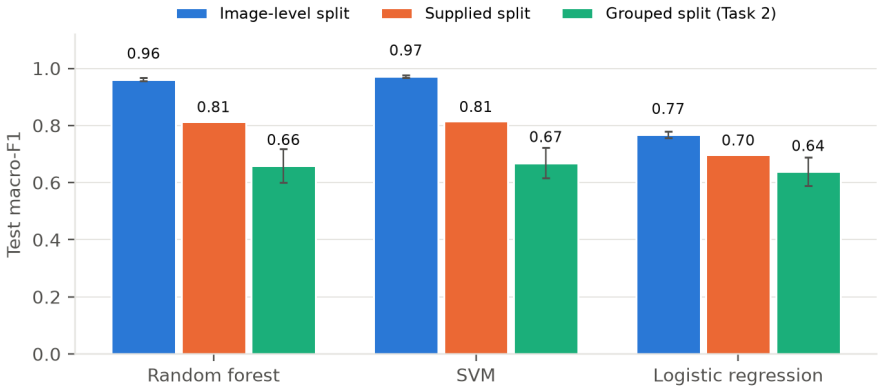
Source	Origin	Images	Healthy	Classes
PK raw	Own field photos, Pakistan	517	120	6
PD raw	Own photos also in PlantDoc	296	17	6
PlantDoc	Web images [6]	469	46	6 + 3
PlantWild	Web, in the wild [8]	1820	217	6 + 2
T.-Village	Field, Rajasthan [7]	1512	216	3 + 2
Nazirpur	Field, Bangladesh [10]	1007	287	2 + bin
Brahmanb.	Field crops, Bangladesh [11]	1035	428	bin
PlantVillage	Laboratory [1]	1098	150	6 + 4
Total		7754	1481	

Classes: six-class labels present + extra disease classes used only in the healthy-vs-diseased task.

4. Results on our own data and single-source transfer (Phases 1-2)

Experiment	Result
Raw set, 10 grouped splits	Macro-F1 RF 0.600, SVM 0.615, LR 0.596; not significantly different.
Robustness (50 perturbations)	Share of clean macro-F1 kept (RF): down-up scaling 0.42, blur 0.43, contrast 0.61, colour cast 0.70, brightness 0.83, rotation 0.96.
Augmentation / illumination	Photometric augmentation helps brightness and contrast robustness; Gray World costs 0.05-0.06 (it shrinks the leaf mask); CLAHE neutral.

Experiment	Result
Source shortcut	Holding out a source cluster lowers accuracy from about 0.61 to 0.47-0.50; metadata alone scores 0.49.
Task 2: supplied 7,200 set	Image-level split 0.961 / 0.972 / 0.768 (RF/SVM/LR); supplied split 0.814 / 0.815 / 0.698; grouped 0.659 / 0.669 / 0.638.
Cross-dataset (trained on raw)	PlantDoc new images 0.29-0.33; Nazirpur 0.21-0.25; Brahmanbaria healthy recall 3-17%. Domain AUC 0.85 (PlantDoc), 0.98-0.99 (Bangladesh).
Explanation transfer	RF importance ranking agreement 0.87 within our data, 0.61 vs PlantDoc, -0.01 vs Nazirpur. Only 2 of 105 features consistent across 4 datasets.



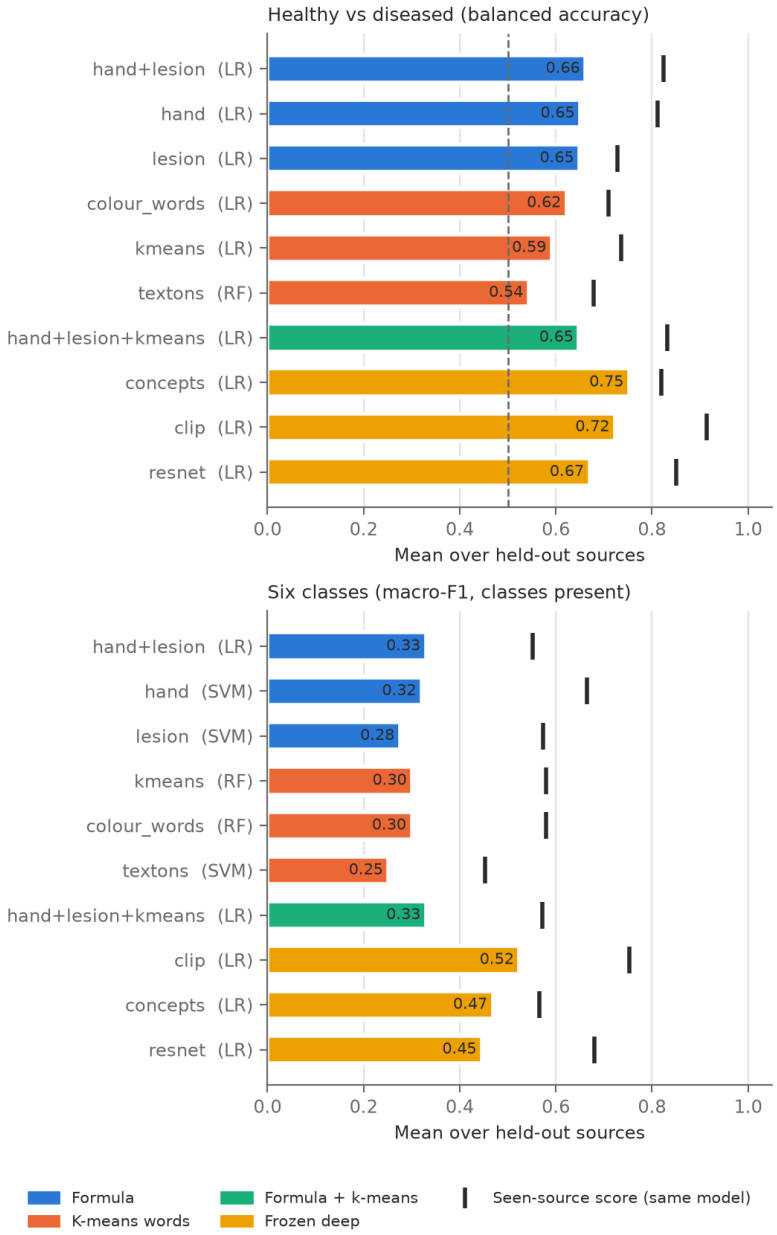
Same images and models, different split rule (Task 2). Error bars: +/- 1 sd over 10 splits.

5. Multi-source evaluation: formula vs k-means vs frozen deep features

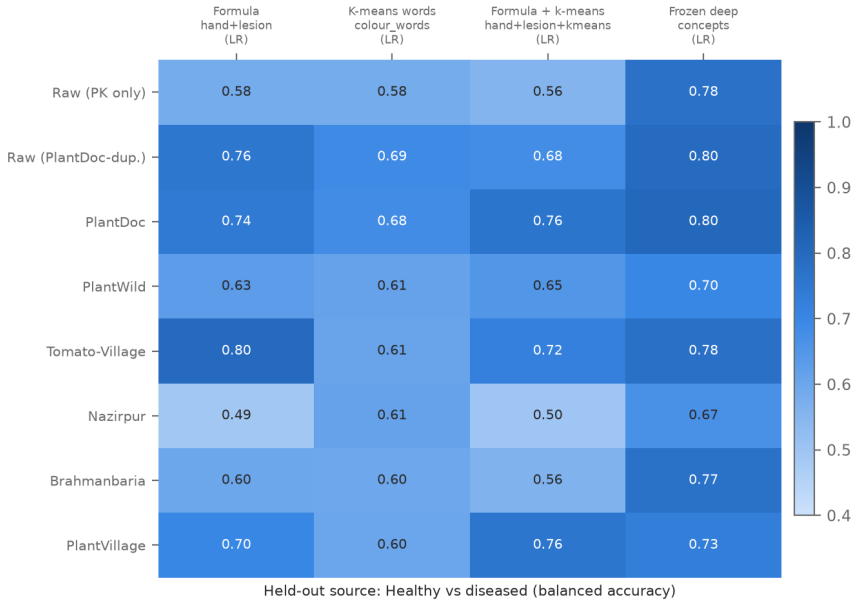
Leave-one-source-out (LOSO): train on seven sources, choose hyper-parameters on 15% of every training source, test on the eighth. Seen-source reference: the same pooled data split 70/15/15 within every source, 5 seeds. Gap = seen minus unseen. All classifiers are classical (RF, RBF SVM, LR); the deep networks are frozen feature extractors.

Features	T1 U	T1 S	T1 gap	T2 U	T2 S	T2 gap
Hand (105)	0.65	0.81	0.16	0.30	0.52	0.23
Lesion (20)	0.65	0.73	0.08	0.25	0.39	0.14
Hand + lesion	0.66	0.82	0.16	0.33	0.55	0.22
Colour words (24)	0.62	0.71	0.09	0.25	0.39	0.14
Textons (32)	0.54	0.61	0.08	0.20	0.31	0.11
Colour words + textons	0.59	0.74	0.14	0.26	0.45	0.19
Hand + lesion + k-means	0.65	0.83	0.18	0.33	0.57	0.24
CLIP concepts (24)	0.75	0.82	0.07	0.47	0.57	0.10
CLIP embedding (512)	0.72	0.91	0.19	0.52	0.75	0.23
ResNet-50 (PCA 256)	0.67	0.85	0.18	0.45	0.68	0.23

Logistic regression, unseen source (U, LOSO) vs seen sources (S), mean over held-out sources; gap = S - U. T1: balanced accuracy, healthy vs diseased (8 sources; chance 0.5). T2: macro-F1 over the classes present (7 sources). Rows 1-3 formula, 4-6 k-means, 7 combined, 8-10 frozen deep. The next figure shows the best of RF, SVM and LR per set.



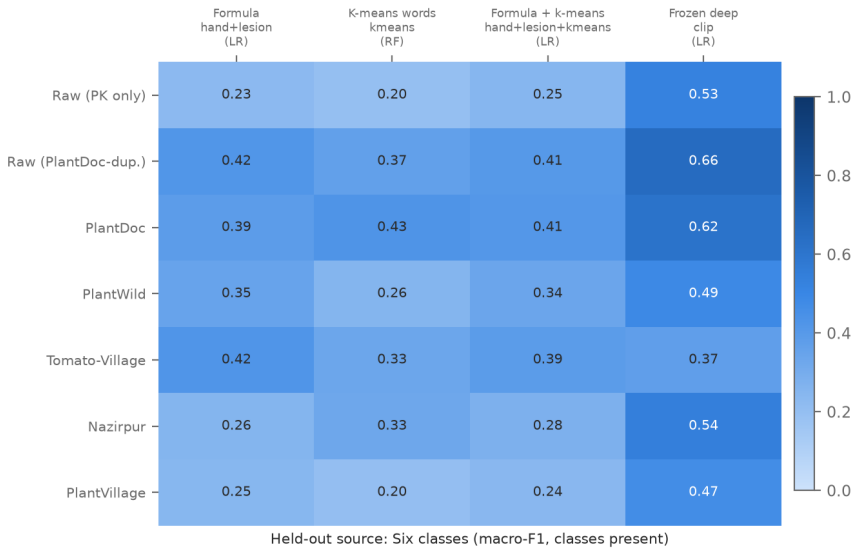
Best classifier per feature set, unseen (bars) vs seen (ticks).



Healthy vs diseased per held-out source, best set of each arm.

Comparison of the two learned-extractor arms

- **K-means (classical):** colour words 0.62 unseen, smallest gap (0.09) and lowest spread across sources; textons add little. Combined with the formula features they do not improve transfer (0.65).
- **Frozen deep:** CLIP concepts 0.75, CLIP embedding 0.72, ResNet-50 0.67 (healthy vs diseased, LR). Six classes: 0.47, 0.52, 0.45.
- **Formula:** hand + lesion 0.66 (binary), 0.33 (six classes, above hand features alone on 7 of 7 sources, not significant after Holm correction).

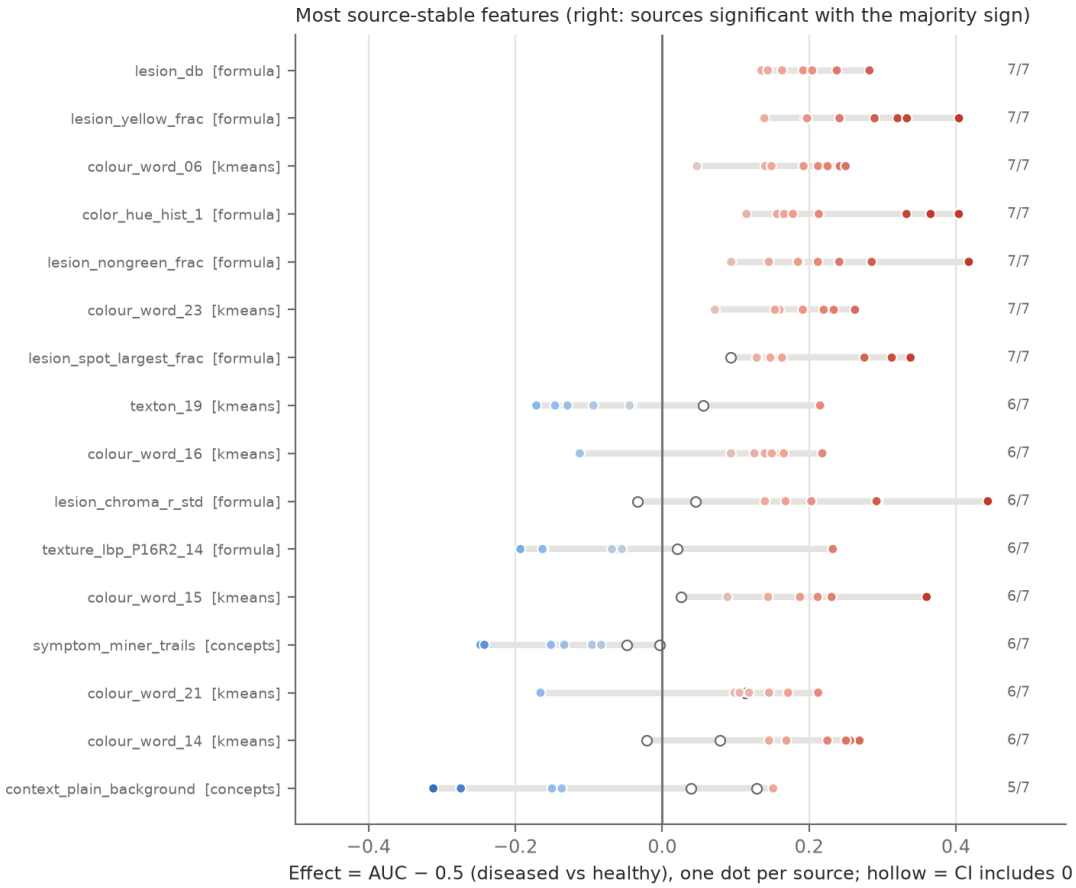


Six classes (macro-F1) per held-out source, best set of each arm.

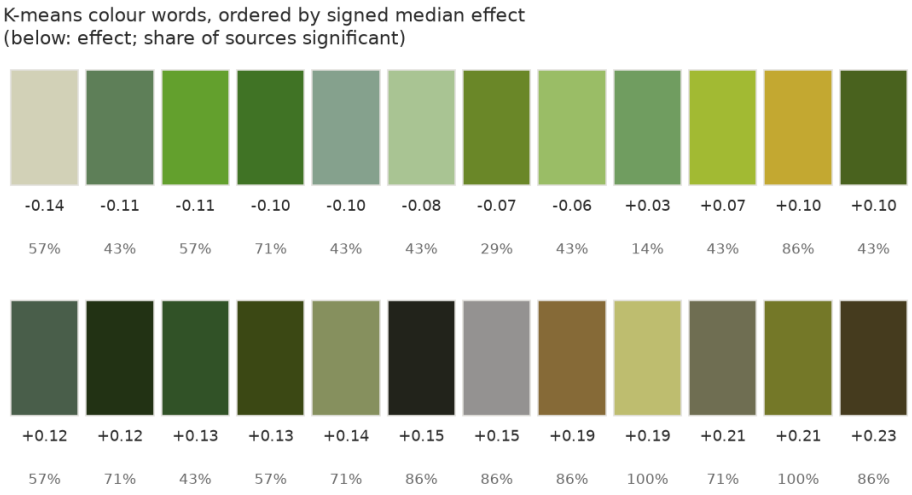
6. Feature-level explanation

Per source with at least 30 healthy and 30 diseased images (7 sources), each feature's effect is AUC - 0.5 for diseased vs healthy with a 1,000-resample bootstrap 95% CI. A feature is source-stable if its CI excludes 0 with the majority sign in at least 75% of sources. No classifier is involved.

Family	Stable	Examples
Hand (formula)	7 / 105	yellow-orange hue share (+), green-red skew (-), 3 LBP bins, GLCM correlation
Lesion (formula)	8 / 20	yellow and non-green fraction, lesion b* shift, largest spot (all +, 7/7 sources)
K-means words	8 / 56	7 brown/olive/ochre colour words (+), 1 texton (-)
CLIP concepts	1 / 24	miner trails (-)



Per-source effects of the most source-stable features.



K-means colour words (learned without labels), ordered by effect.

Of 205 named features, 24 are source-stable. Leaf-relative lesion features are the most reliable family (8 of 20): the yellow and non-green fractions of the leaf, the lesion-minus-tissue shift in b* and the size of the largest spot are higher in diseased leaves on all seven eligible sources. Of the 105 hand features only 7 qualify, chiefly the yellow-orange hue share. The k-means codebook, learned without labels, isolates the same signal in a form that can be shown rather than defined: its stable words are browns, olives, ochres and near-black, while the green words have negative effects that are not stable. Only one CLIP concept is stable.

Model-based explanations do not recover this set. The Spearman correlation between RF permutation importance on the held-out source and feature stability ranges from -0.22 to 0.16 across folds. The L1-LR on CLIP concepts, the strongest unseen-source model, keeps sign-stable weights on 4 context concepts in every LOSO fold - plain background (towards healthy), in hand and studio (towards diseased) - alongside symptom concepts such as necrotic edges and target spots. Part of its advantage comes from capture context that happens to be shared between sources, not from symptoms.

Pool	L1-LR all	L1-LR stable	RF all	RF stable
Formula	0.64	0.63	0.58	0.57
K-means	0.60	0.61	0.56	0.56
Formula + k-means	0.64	0.62	0.56	0.58
CLIP concepts	0.76	0.57	0.66	0.51

Explanation-guided selection: LOSO balanced accuracy with all features vs only the features found source-stable on the training sources.

Restricting models to stable features does not improve transfer: formula L1-LR 0.64 to 0.63, k-means 0.60 to 0.61, and CLIP concepts lose 0.18 because the context concepts carry transferable signal. Stability identifies which features mean the same thing everywhere; the other features still add predictive, if less interpretable, information.

7. Answers to the research questions

Question	Answer
RQ1 Accuracy of classical ML	About 0.60 macro-F1 within our data; on unseen sources 0.66-0.75 balanced accuracy (binary); six classes 0.33 macro-F1 with formula or k-means features, 0.52 with frozen CLIP embeddings.
RQ2 Robustness	Resolution and blur hurt most; photometric augmentation helps lighting only. The largest 'perturbation' is a change of source.
RQ3 Shortcuts and leakage	Strong evidence: related images inflate scores by 0.3; a third of our data is PlantDoc; source clusters and capture-context concepts carry signal.
RQ4 Useful, interpretable, stable features	24 source-stable features, mainly leaf-relative yellow/brown/non-green fractions, lesion size and brown/olive colour words. They are interpretable and stable but, used alone, do not raise unseen-source accuracy.

8. Limitations

- Labels come from different annotators; 72 cross-source conflicts are recorded, not resolved.
- 'Diseased' mixes different diseases per source; explanation covers healthy vs diseased only.
- The HSV leaf mask is heuristic; deep extractors see the whole image, which favours context cues.
- Only frozen, generic deep extractors; no fine-tuning (by design: classical ML only).
- Overlap with PlantDoc: 296 of 813 raw photos by the content rule here vs 263 of 830 reported earlier against six-class PlantDoc images; not yet reconciled image by image.
- Runs were on a 4-core CPU machine; RF results differ slightly from the original machine (in-domain 0.613 vs 0.600).

9. Next steps

- Label review of the 72 cross-source conflicts and the 21 septoria/early-blight suspects.
- Per-disease explanation (disease vs healthy for each disease) on sources that have it.
- Leaf masking before the deep extractors, to test how much of their advantage is context.
- Team: read the cited papers before submission (course rule: cite only sources read).

10. Reproducibility

- Code: scripts/24 (sources) - 25/25b (features) - 26 (LOSO) - 27 (explanation) - 28 (summary) - 29 (figures); src/features_lesion.py, src/codebook.py; configs/sources.yaml, configs/concepts.yaml.
- Environments: .venv (Python 3.11.9, requirements.txt); .venv-dl for the frozen extractors (requirements-dl.txt).
- Outputs: F:/NUST/ML/runs/p4_multisource (results/, logs/). Numbers: docs/verified_results.md §§11-15. Commands: docs/experiment_log.md. Design: docs/phase3_design_2026-11-12.md.